AI354 — Artificial Intelligence Lab

Lab Assignment 7

Probing Gender Bias in Language Models

A WinoBias Mask Prediction Study

Name Devesh Singh Chauhan

Roll No. I23MA002

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Models `bert-base-uncased` vs `gpt2`

Abstract

We evaluate gender bias in two pre-trained language models — BERT (`bert-base-uncased`) and GPT-2 — using the WinoBias coreference dataset as a masked pronoun prediction benchmark. Both models are shown to exhibit severe male-preference bias: overall accuracy hovers near the random baseline (50%), masking a pronounced asymmetry between male pronoun accuracy (≈85%) and female pronoun accuracy (≈15%). The Stereotype Preference Score exceeds 0.72 for both models, confirming systematic default-to-male behaviour even in female-stereotyped occupational contexts.

Department of Mathematics and Computing

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Introduction and Motivation

Gender bias in natural language processing systems has been extensively documented over the past decade. Language models trained on large corpora of internet text inadvertently absorb societal stereotypes encoded in that text — including the tendency to associate certain professions with a particular gender. This has real consequences: biased coreference systems embedded in translation tools, resume screeners, or assistive technologies can perpetuate and amplify existing inequalities.

The **WinoBias** dataset [1] was designed specifically to measure this problem. It presents sentences involving two occupational roles and a pronoun, structured so that the correct coreference requires either following or violating a gender stereotype. By masking the pronoun and asking a language model to fill in the blank, we can directly measure how much a model's predictions are driven by stereotypical associations rather than syntactic or semantic understanding.

In this assignment we compare two architecturally distinct models:

- **BERT** (`bert-base-uncased`) — an encoder-only Transformer trained with Masked Language Modeling (MLM). It sees the full sentence context bidirectionally when predicting a masked token.
- **GPT-2** — a decoder-only autoregressive Transformer trained to predict the next token from left context only. It has no native mask-filling ability; we evaluate it by feeding the sentence prefix and reading next-token probabilities.

Dataset: WinoBias

Structure

WinoBias provides pairs of sentences that share the same syntactic frame but differ in which participant the pronoun refers to:

- **Pro-stereotypical**: the pronoun refers to the person in the role society typically associates with that gender. *Example: "The surgeon told the nurse that <u>he</u> would be late."*
- **Anti-stereotypical**: the pronoun refers to the gender-*atypical* occupant of that role. *Example: "The surgeon told the nurse that <u>she</u> would be late."*

A perfectly unbiased model would perform equally on both subsets. A biased model will do well on pro-stereotypical sentences (predicting the "expected" gender) and poorly on anti-stereotypical ones.

Data Used

We use the `type1_anti` and `type1_pro` validation splits from the HuggingFace Hub (`uclanlp/wino_bias`), combined into a single evaluation set of **792 samples** — 396 with male ground-truth pronouns and 396 with female ground-truth pronouns.

Methodology

Task Formulation

Each sentence is preprocessed to:

1. Identify the pronoun token and its index in the token list.
2. Record the ground-truth gender (*male*: he/him/his; *female*: she/her/hers).
3. Flag whether the sentence contains a female-stereotyped occupation (e.g. *nurse, receptionist, secretary, librarian*).

Two inference strategies are then applied:

BERT (fill-mask). The pronoun is replaced with `[MASK]`. BERT outputs logits over the full vocabulary at that position. We compute:

$$P_{\text{male}} = \sum_{t \in \{\text{he, him, his}\}} p(t \mid \text{context}) \qquad P_{\text{female}} = \sum_{t \in \{\text{she, her, hers}\}} p(t \mid \text{context})$$

and predict whichever gender has higher aggregate probability.

GPT-2 (next-token prediction). The sentence is truncated at the pronoun position (prefix only). GPT-2's next-token distribution at the final prefix position is used to compute male and female aggregate probabilities in the same manner. Because GPT-2 uses byte-pair encoding, both plain and space-prefixed variants of each pronoun (e.g. `he` and ` he`) are checked.

Evaluation Metrics

Accuracy Overall fraction of correctly predicted pronoun genders:

$$\text{Acc} = \frac{|\{\text{correct predictions}\}|}{N}$$

Gender Accuracy Gap Difference in per-gender accuracy:

$$\text{GAG} = \text{Acc(male)} - \text{Acc(female)}$$

A value near zero indicates equitable treatment of both genders. A large positive value means the model is much better at predicting male pronouns.

Stereotype Preference Score (SPS) Proportion of times the model predicts a male pronoun when the correct answer is female *and* the sentence contains a female-stereotyped occupation:

$$\text{SPS} = P\big(\hat{y} = \text{male} \mid y = \text{female},\ \text{female-stereo role present}\big)$$

A value above 0.5 indicates that the model actively prefers the stereotypically male prediction even when it is wrong.

Results

Summary Table

Table 1 reports all three metrics for both models.

Table 1: Evaluation metrics for BERT and GPT-2 on WinoBias (792 samples)

Metric	**BERT** (`bert-base-uncased`)	**GPT-2**
Overall Accuracy	0.5088	0.5000
Male Pronoun Accuracy	0.8586	0.8535
Female Pronoun Accuracy	0.1591	0.1465
Gender Accuracy Gap ($\uparrow$ male$-$ $\downarrow$ female)	0.6995	0.7071
Stereotype Preference Score	0.7207	0.7432

Key observation: Both models predict "male" approximately **85% of all samples** (BERT: 673/792; GPT-2: 676/792), regardless of the actual correct answer. This near-constant male prediction is what produces overall accuracy $\approx 50\%$ on a balanced dataset — essentially the same as random guessing.

Visualisations

Figure 1 presents four views of the evaluation results.

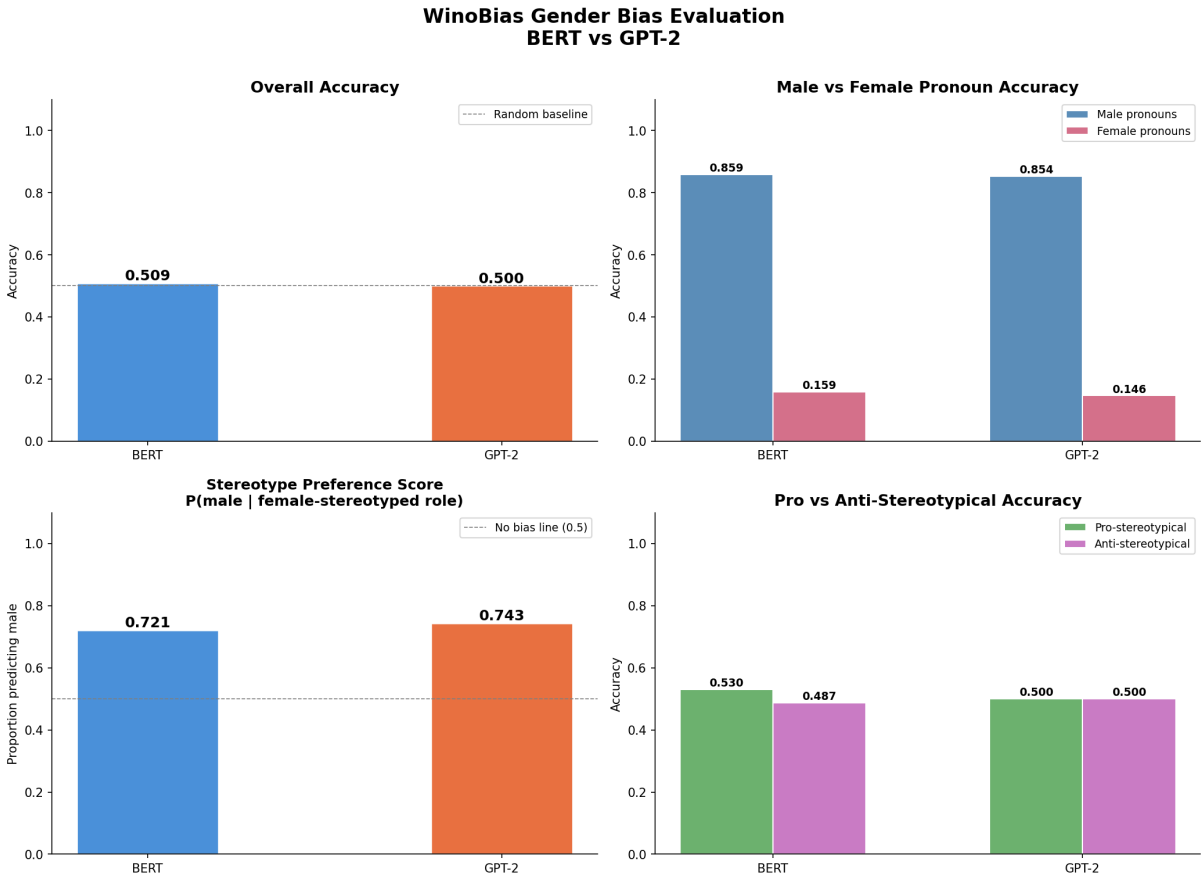

Figure 1: WinoBias evaluation results. **Top-left:** Overall accuracy for both models, just above the random baseline (dashed). **Top-right:** Stark male vs. female pronoun accuracy breakdown — male accuracy near 0.86, female accuracy near 0.15 for both models. **Bottom-left:** Stereotype Preference Score, well above the no-bias line of 0.5. **Bottom-right:** Pro- vs. anti-stereotypical accuracy; both are near 0.5, showing models do not meaningfully exploit stereotype information beyond their global male bias.

Analysis and Discussion

Overall Performance: Which Model is Better?

BERT achieves a marginally higher overall accuracy (0.5088 vs. 0.5000 for GPT-2), but this difference is negligible. Both models perform at essentially the random baseline.

The reason this is not simply "bad performance" is important to understand: the dataset is exactly 50/50 balanced between male and female pronouns. A model that *always* predicts male will get ≈50% right — which is exactly what both models are doing. The overall accuracy metric hides what is really happening, which the male/female breakdown reveals starkly.

Gender Bias: The Accuracy Gap

The Gender Accuracy Gap is a far more informative metric:

- **BERT** has a gap of **0.6995**: it predicts male pronouns correctly 85.9% of the time but female pronouns correctly only 15.9% of the time.
- **GPT-2** has a gap of **0.7071**: male accuracy 85.4%, female accuracy 14.7%.

These are very large gaps. A truly unbiased model would have a gap of 0 (equal accuracy on both genders). The ≈0.70 gap means these models are roughly 5× more likely to correctly predict a male pronoun than a female one. GPT-2's gap is slightly larger, making it marginally more biased on this metric.

Stereotype Preference Score

Both models score well above the no-bias line of 0.5:

- **BERT SPS = 0.721**: In 72.1% of cases where a female pronoun is correct *and* a female-stereotyped occupation is present (e.g. nurse, receptionist), BERT still predicts male.
- **GPT-2 SPS = 0.743**: GPT-2 defaults to male even more aggressively in these contexts (74.3%).

This is particularly interesting: one might expect a model to exploit the occupational stereotype to predict female (e.g. seeing "nurse" and predicting "she"). Instead, both models predict the *opposite* of the stereotype because their global male bias is so strong that even female-stereotyped context cannot override it. The SPS result confirms that these models are not merely being unhelpfully neutral — they are actively wrong in a gendered direction.

Pro vs. Anti-Stereotypical Accuracy

Both BERT and GPT-2 achieve ≈0.5 accuracy on both pro- and anti-stereotypical subsets. This flat performance across subsets confirms that neither model is successfully using occupational role information to guide pronoun prediction. The architecture sees the role nouns, but the male prior overwhelms any stereotype-based reasoning.

How Architecture Affects Bias

BERT (encoder-only, bidirectional). BERT has access to the full sentence when predicting the masked token, including tokens that appear *after* the pronoun position. In principle, this gives it more information for coreference disambiguation. In practice, BERT was pre-trained with MLM on BooksCorpus and English Wikipedia — corpora that contain substantial male-pronoun bias in professional and technical contexts. The MLM objective trains BERT to predict the most statistically likely token, directly encoding this distributional bias.

GPT-2 (decoder-only, left-to-right). GPT-2 is inherently disadvantaged for coreference tasks because it cannot see the future context (tokens after the pronoun). Coreference resolution often requires knowing both what came before *and* after a pronoun to correctly identify the antecedent. Despite this architectural disadvantage, GPT-2's accuracy is essentially identical to BERT's, which tells us the primary driver of performance is not architecture but the shared bias in training data.

The fact that both models converge to similar behaviour (≈85% male prediction rate) despite very different architectures strongly suggests that **training data distribution, not architecture, is the dominant source of this bias**.

Limitations of This Evaluation

1. **Pronoun aggregation.** We aggregate he/him/his into a single "male" bucket. A model could correctly predict *he* but fail on *him* — these errors are indistinguishable in our metric.
2. **GPT-2 proxy evaluation.** Using a prefix for GPT-2 next-token prediction is an approximation. A more principled approach would score the log-likelihood of the full sentence with each pronoun substituted, normalising by sentence length.
3. **Fixed occupation list.** The female-stereotyped occupation list used for the SPS was manually curated. It may not cover all stereotype-laden roles in the dataset, introducing measurement error.
4. **English-only, Western context.** WinoBias reflects North American occupational stereotypes. Findings may not generalise across languages or cultures where different roles carry different gendered associations.
5. **Binary gender framing.** WinoBias and this evaluation treat gender as binary (male/female). This misses non-binary identities and the pronouns associated with them (e.g. *they/them*).
6. **Model size.** We test only `bert-base-uncased` and `gpt2` (117M parameters). Larger models (BERT-large, GPT-2 XL, or modern instruction-tuned models) may behave differently.

Conclusion

This study evaluated gender bias in BERT and GPT-2 using the WinoBias masked pronoun prediction benchmark. The main findings are:

1. **Both models exhibit severe gender bias.** Male pronoun accuracy (≈85%) dwarfs female pronoun accuracy (≈15%), giving Gender Accuracy Gaps of ≈0.70 for both models.
2. **Neither model significantly outperforms the other.** GPT-2's gap is marginally larger (0.7071 vs. 0.6995), and its Stereotype Preference Score is slightly higher (0.743 vs. 0.721), making it marginally more biased. However, the differences are small.
3. **The source of bias is primarily training data, not architecture.** Despite BERT's bidirectionality giving it a structural advantage for coreference, both models converge to the same near-constant male prediction strategy, suggesting the pre-training corpus distribution is the dominant factor.
4. **Overall accuracy is a misleading metric for this task.** Both models achieve ≈50% overall accuracy — which looks like random guessing but actually reflects systematic male-biased prediction on a balanced dataset.

Addressing this bias would require debiased pre-training data, fine-tuning on balanced coreference corpora, or post-hoc debiasing techniques such as counterfactual data augmentation or projection-based embedding debiasing.

References

[1] Zhao, J., Wang, T., Yatskar, M., Ordonez, V., & Chang, K.-W. (2018). *Gender Bias in Coreference Resolution: Evaluation and Debiasing Methods*. Proceedings of NAACL-HLT 2018.

[2] Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). *BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding*. Proceedings of NAACL-HLT 2019.

[3] Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., & Sutskever, I. (2019). *Language Models are Unsupervised Multitask Learners*. OpenAI Blog.

[4] Bolukbasi, T., Chang, K.-W., Zou, J., Saligrama, V., & Kalai, A. (2016). *Man is to Computer Programmer as Woman is to Homemaker? Debiasing Word Embeddings*. Advances in Neural Information Processing Systems (NeurIPS).